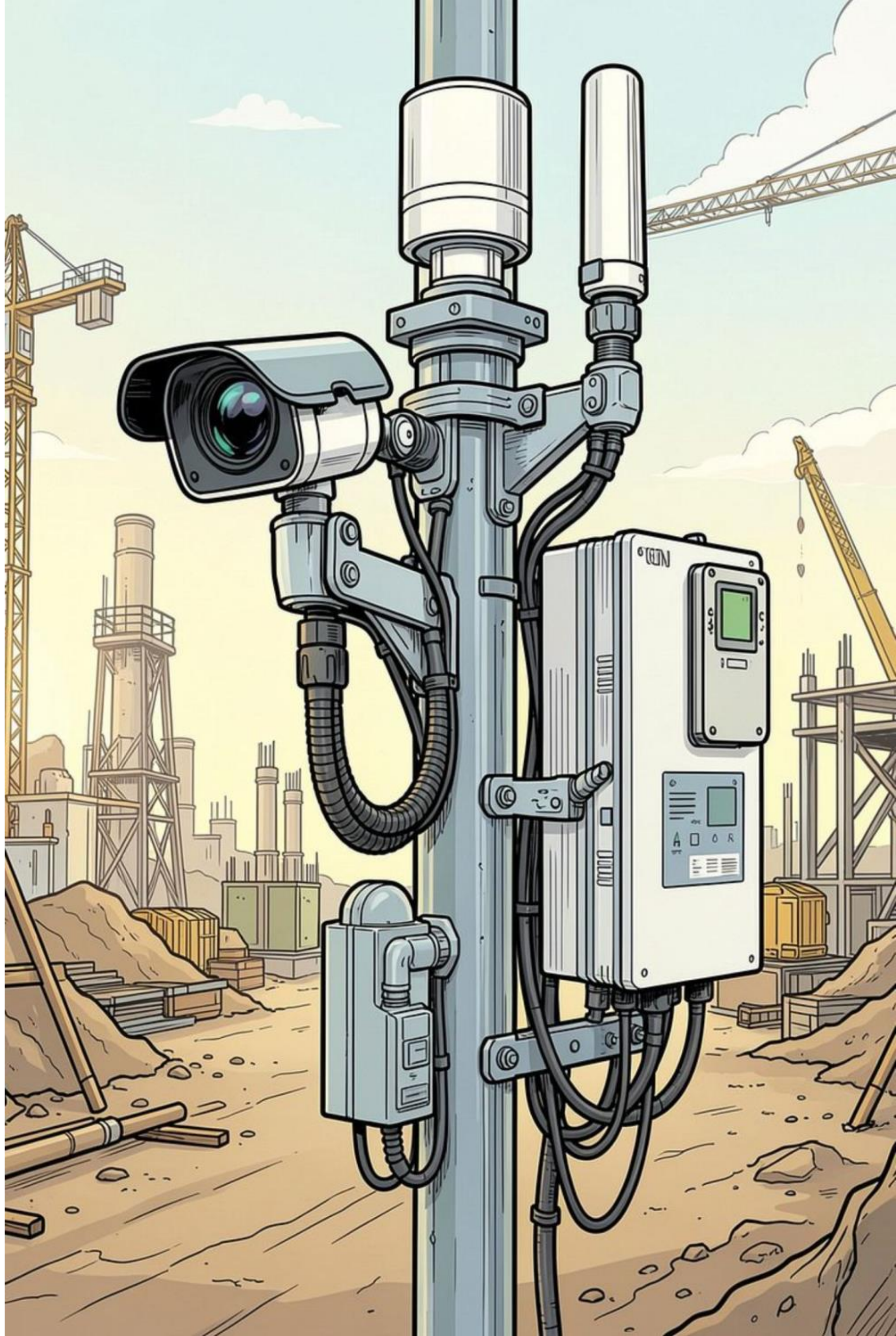
Software Engineering Laboratory

AI-Enabled IoT-Based Real-Time Environmental and Surrounding Safety Monitoring System

SmartSurround

Intelligent Monitoring · Detection · Prediction · Safety



The Problem: Reactive Monitoring Isn't Enough

Traditional systems monitor one or two parameters, require manual observation, and only warn *after* unsafe conditions occur — leaving workers and the public exposed.

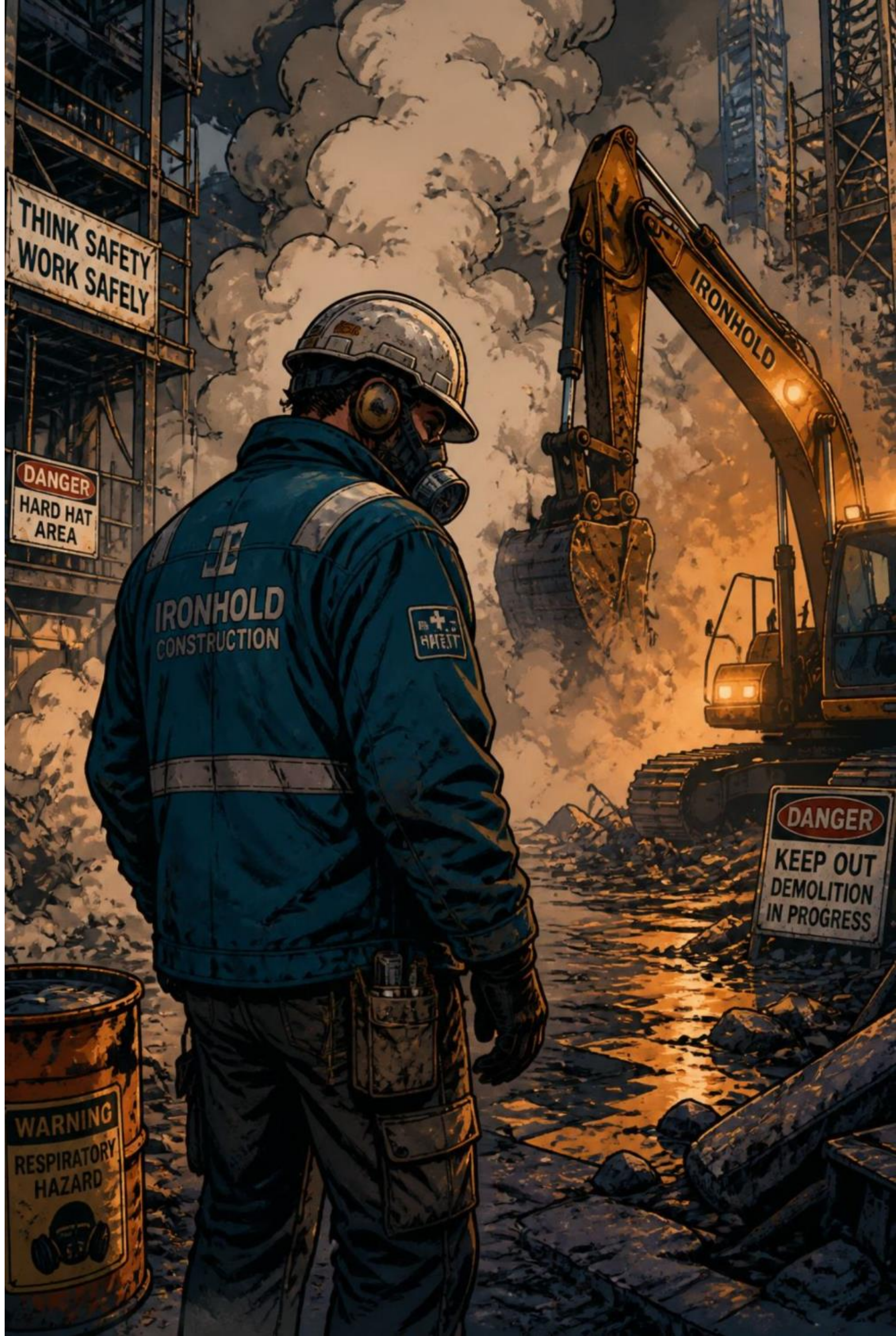
❌ Traditional Approach

Manual checks → Delayed detection → Risk exposure → Reactive response

✅ SmartSurround

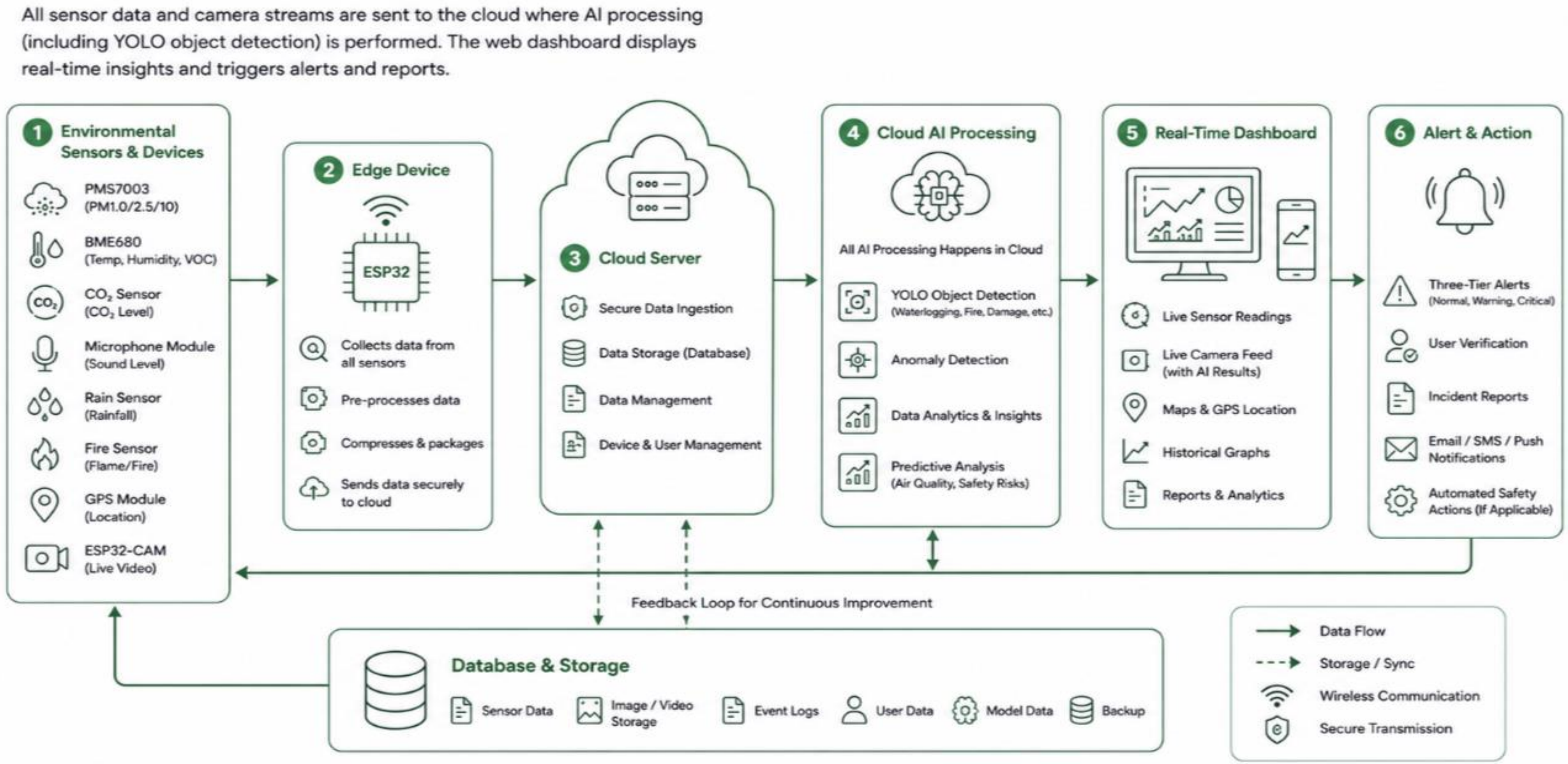
Continuous sensing → AI detection → Prediction → Proactive safety action

Core motivation: The system should **understand** the environment, identify risks, predict problems, and help people act — not just collect data.



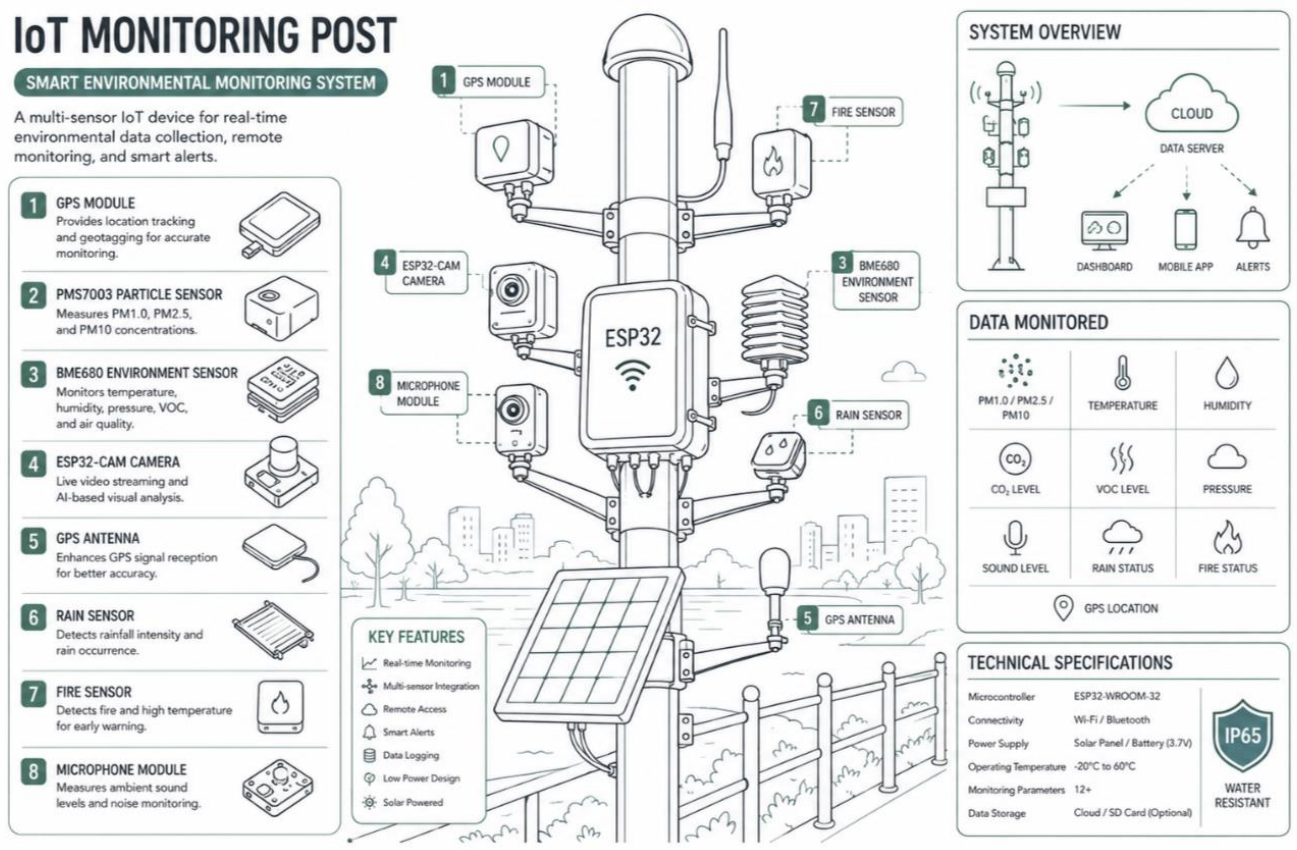
System Architecture & Working Flow

Sensors collect data, the ESP32 transmits it wirelessly, and a web dashboard displays real-time readings with AI analysis triggering alerts and incident reports.



The ESP32-CAM also enables AI-powered visual anomaly detection — identifying fire, waterlogging, damaged roads, and other hazards from live camera imagery.

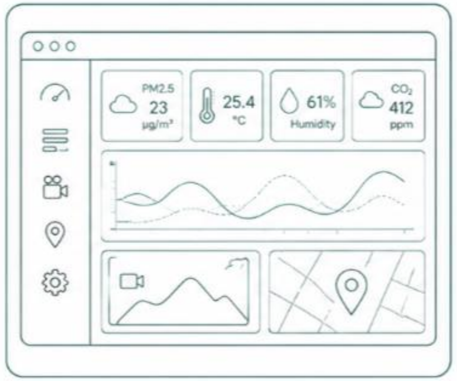
Hardware Components & Sensing Parameters

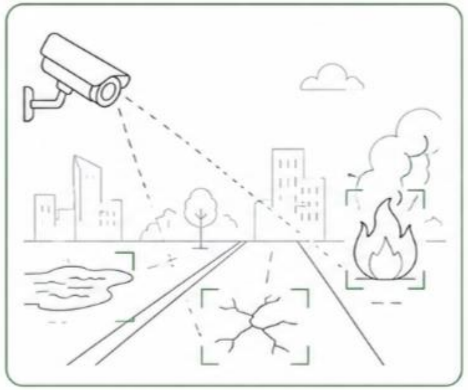


ESP32	Main IoT controller & wireless comms
ESP32-CAM	Live video & AI visual detection
PMS7003	PM1.0, PM2.5, PM10
BME680	Temperature, humidity, VOC
CO_2 Sensor	Carbon dioxide levels
Mic Module	Sound-level monitoring
Rain Sensor	Rain detection
Fire Sensor	Flame detection
GPS Module	Lat, long & location tracking

All components are mounted on a single monitoring post, continuously observing the surrounding environment.

AI-Powered Monitoring & Intelligent Web Dashboard





1 Real-Time Dashboard

Live readings for all sensors, GPS coordinates, and camera feed. Historical graphs track trends and flag anomalies.

2 AI Camera Analysis

ESP32-CAM detects waterlogging, damaged roads, and fire/smoke. Detections require **user verification** before generating reports — reducing false alerts.

3 Intelligent Alert System

Three-tier alerts: **Normal → Warning → Critical**. Triggers dashboard warnings, notifications, and incident reports when thresholds are crossed.

Applications & Intelligent Safety Management

Role-Based Access

- **Admin:** Full dashboard access, threshold configuration, AI monitoring, reports
- **Worker:** Real-time conditions and safety warnings — no threshold modification

Predictive Safety Response

Historical data trains AI/ML models to predict risk before it escalates.

Example: Rising dust + temperature → early warning → automated sprinkler/dust suppression → continued monitoring. Automated control uses only safety-rated hardware and predefined rules.

Expected Outcomes, Future Scope & Conclusion

Expected Outcomes

- Continuous real-time environmental monitoring
- AI-assisted visual anomaly detection
- GPS-based incident reporting with evidence
- Early warning and predictive analysis
- Role-based access and daily safety reports

Future Scope

- Advanced AI models and edge AI deployment
- Multi-post control center and mobile app
- Smart-city integration and cloud deployment
- Digital map of all posts and incidents

> "SmartSurround transforms environmental monitoring from simple data collection into intelligent sensing, detection, prediction, and safety management."

Thank You

Questions & Discussion